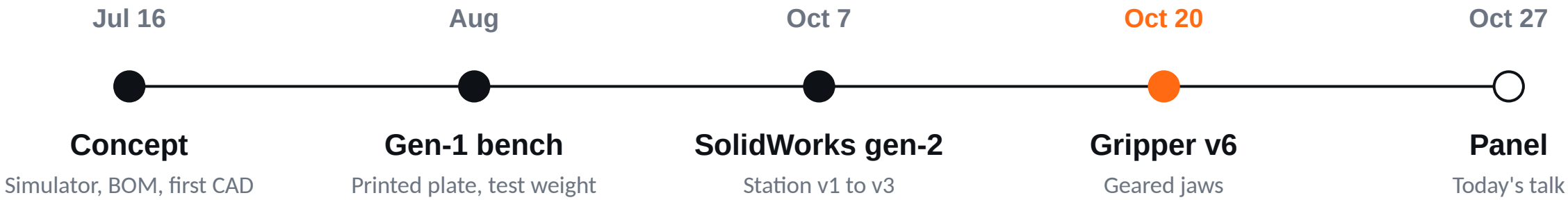
RoomCleaner

A four-cable ceiling robot
that picks laundry off the floor

Jasper Buntinx · Technical presentation · Oct 27, 2026



Solo build: four winch stations, a camera claw, and the software.



BOUGHT

A4988 driver, ESP32, DROK buck, MG996R servo.

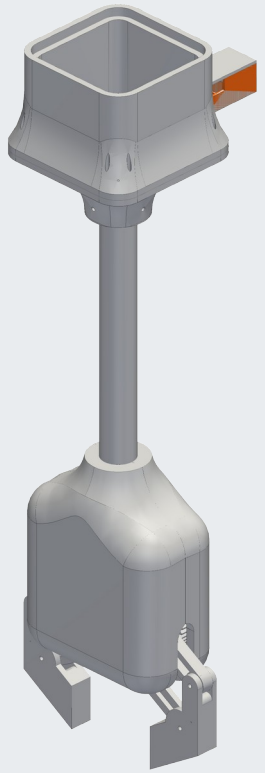
MADE BY ME

Everything else: designed, printed, machined and wired.

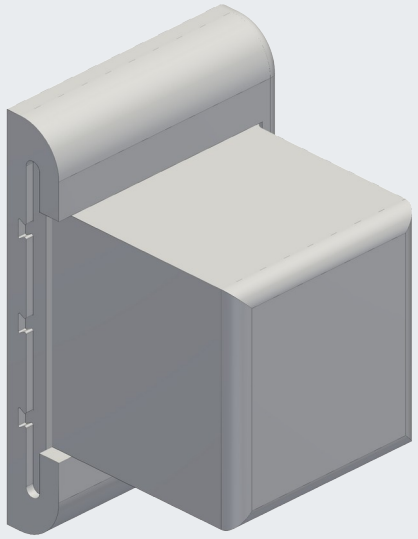
HOW TO READ THE TAGS

MEASURED	On hardware, with an instrument
CALCULATED	Hand calculation, stated inputs
SIMULATED	Run in a model or simulator
PLANNED	Not done yet

Done means jeans from anywhere on the floor into the hamper.



Claw v6 with camera bay

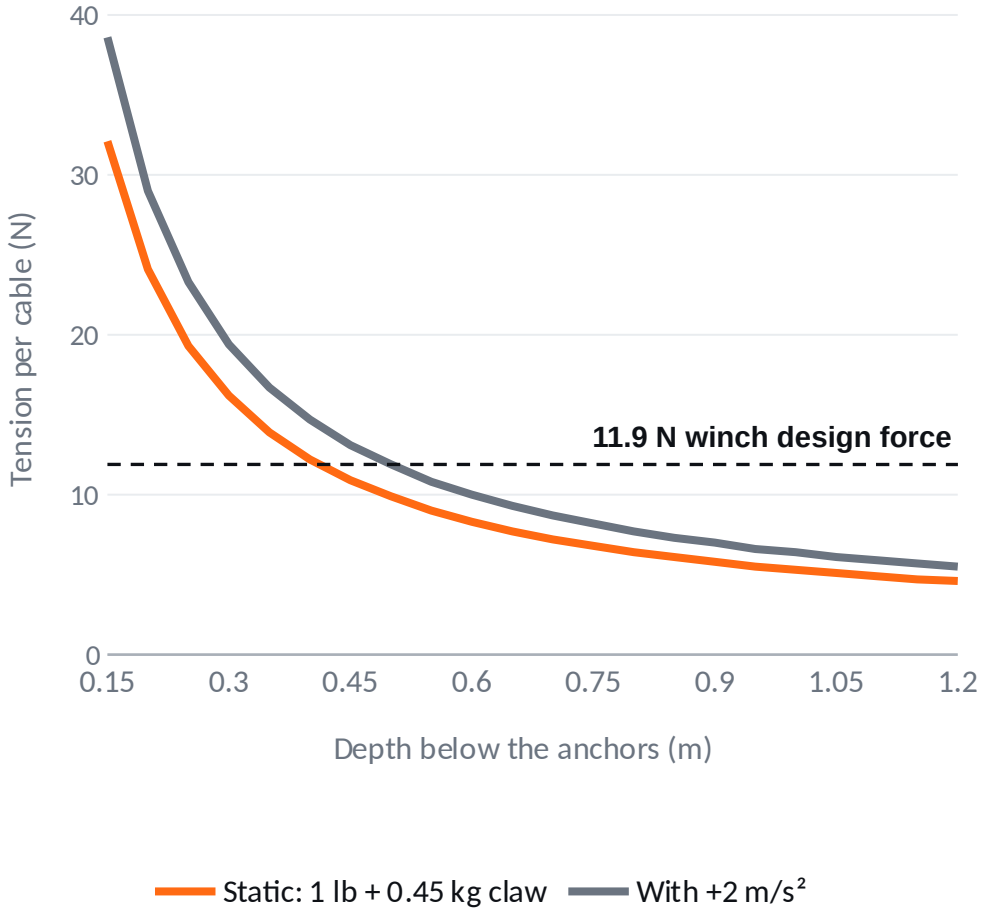


Winch station v3

0.9 kg	Payload of jeans, plus 2 m/s² acceleration
≤ 450 g	Claw mass budget
4 × studs	Winch stations screwed into wall studs
1 switch	The switched power strip is the only kill switch
0 people	Nobody under the robot while it moves

CALCULATED

Cable angle, not payload, sets the limits; walls bind before the ceiling.



1	WANTED	Use the whole room.
2	REALIZED	**32 N at 0.15 m depth, 8.3 N at 0.6 m.**
3	DECIDED	Ceiling margin 0.4 m, plus a wall margin.
4	RESULT	At task heights, corners bind (1.1 × weight).
5	BUT	Claw mass counts like payload.

FUNDAMENTALS

$$T = W / (4 \sin \varphi),\ \sin \varphi = d / \sqrt{(d^2 + D_h^{\,2})}$$

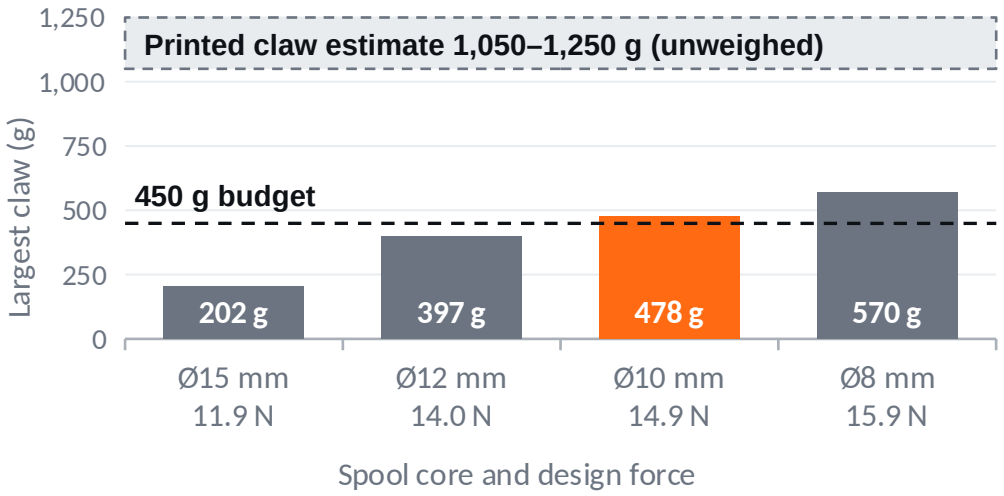
$W = 8.86$ N (1 lb + 0.45 kg claw), $D_h = 2.17$ m: $T = 32.1$ N at $d = 0.15$ m, 8.3 N at 0.60 m.

MEASURED
CALCULATED

The driver, not the motor, capped my pull; geometry buys it back.

0.85 V → 1.06 A

Vref cap on my A4988 boards, and the current it allows



PLANNED — result after the Oct 22 test

Luggage-scale pull at 1.06 A, empty and full spool; replaces the assumed 0.7 running-torque factor.

1 **WANTED** Raise current from 1.0 to 1.35 A.

2 **REALIZED** **Vref pot maxes at 0.85 V: 1.06 A cap.**

3 **DECIDED** Run at the cap; shrink the spool core.

4 **RESULT** Winches hold ~1.1 kg; claw plus jeans ~2 kg.

5 **BUT** Torque factor 0.7 assumed; pull test decides.

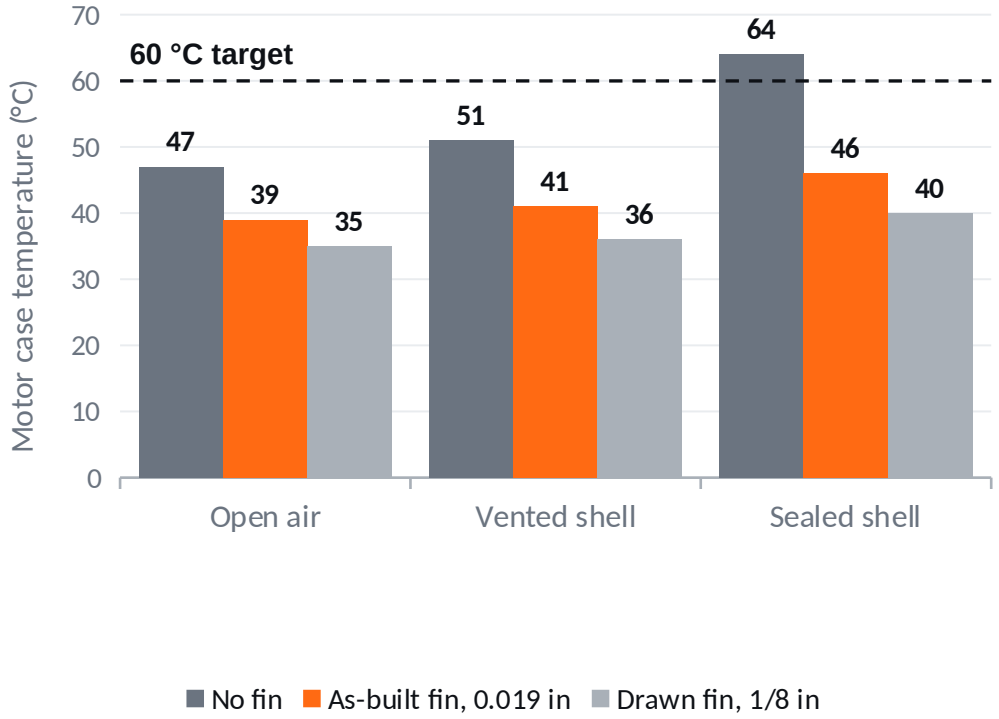
FUNDAMENTALS

$$I = V_{ref} / (8 \cdot R_s) = 0.85\ V / 0.8\ \Omega = 1.06\ A$$

$T = \tau / r$: $\tau_{run} = 0.7 \times 0.255\ N{\cdot}m = 0.179\ N{\cdot}m$, so $T = 23.8\ N$ at $r = 7.5\ mm$ and $17.9\ N$ at a full spool ($r = 10\ mm$).

CALCULATED

Sealing the station made the motor mount its hottest, most loaded joint.



PLANNED — result after the Oct 22 test

Hold test at 1.06 A with thermometers on the motor case, fin and PETG tab; pass limit below 60 °C.

1	WANTED	A clean, closed printed shell.
2	REALIZED	Holding stepper heats (I^2R); PETG conducts 1/800 of aluminium.
3	DECIDED	Vents plus an aluminium fin.
4	RESULT	**Sealed, 1.06 A: 64 °C bare, 46 °C finned.**
5	BUT	No thermometer data yet; hold test planned.

FUNDAMENTALS

$$P = I^2R;\quad R_{cond} = L / (k\,A);\quad R_{conv} = 1 / (h\,A)$$

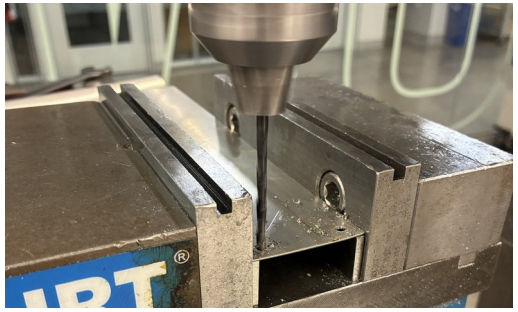
$P = 1.06^2\ A^2 \times 1.8\ \Omega = 2.0$ W (R assumed). Flange path: PETG 14 K/W vs aluminium 0.011 K/W, 1260 × worse.

MEASURED
CALCULATED

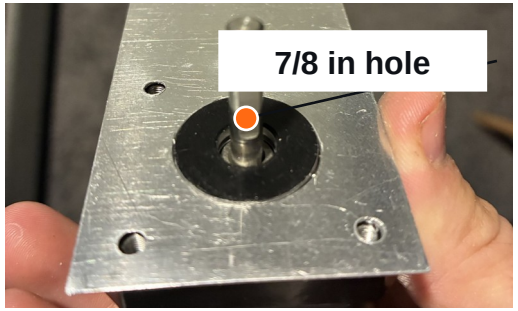
I drew 1/8-inch, built 0.019-inch, then re-checked the physics.



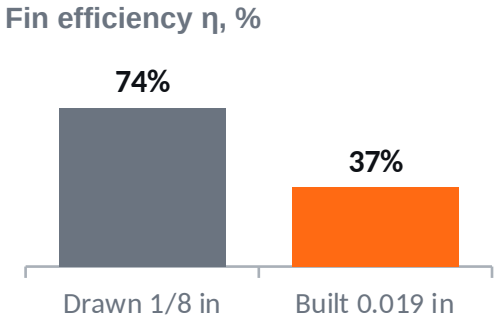
1 Sheared blanks beside my drawing



2 Drilling the holes on the mill



3 Fitted: boss centred, 4 holes line up



PLANNED — result after the Oct 22 test

Hold test: does the thin sheet bow or buzz, and are the screw threads deep enough?

1 **WANTED** Drawn: 1/8 in plate, fin efficiency 74 %.

2 **REALIZED** The sheet I could drill was 0.019 in.

3 **DECIDED** **Re-ran the fin equation: 74 % falls to 37 %.**

4 **RESULT** Still carries about half the heat: 46 °C.

5 **BUT** May buzz; screws reach 2.7 mm deeper.

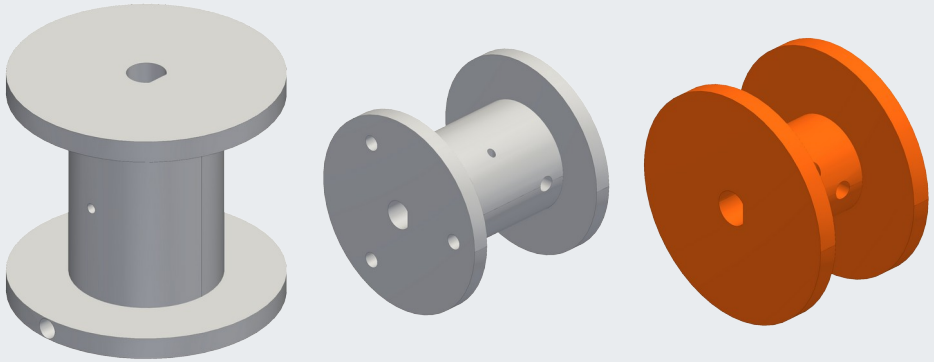
FUNDAMENTALS

$$\eta = \tanh(mL) / (mL), \quad m = \sqrt{(2h / (k\,t))}$$

t = 0.483 mm: m = 13.5 /m, mL = 2.69, η = 37 %, R_{fin} = 22 K/W. Drawn 1/8 in: mL = 1.05, η = 74 %, R_{fin} = 10.9 K/W.

CALCULATED

My first spool outran its motor shaft; the fix changed the math.



Gen-1 spool
Ø20 core, 32 mm long

v2
bore Ø5.2, too tight

v3
Ø15 core, bore Ø5.45

Ø20 → Ø15
core, mm

32 → 22
overall length, mm

Ø5.2 → Ø5.45
bore, mm

1 **WANTED** — 32 mm drum on a Ø20 core.

2 **REALIZED** — Stuck out 14.5 mm past the 24 mm shaft.

3 **DECIDED** — **Ø15 core, 16 mm drum; radius tracks wound length.**

4 **RESULT** — v2 bore too tight: Ø5.2 opened to Ø5.45.

5 **BUT** — Printed holes shrink; bore never calipered.

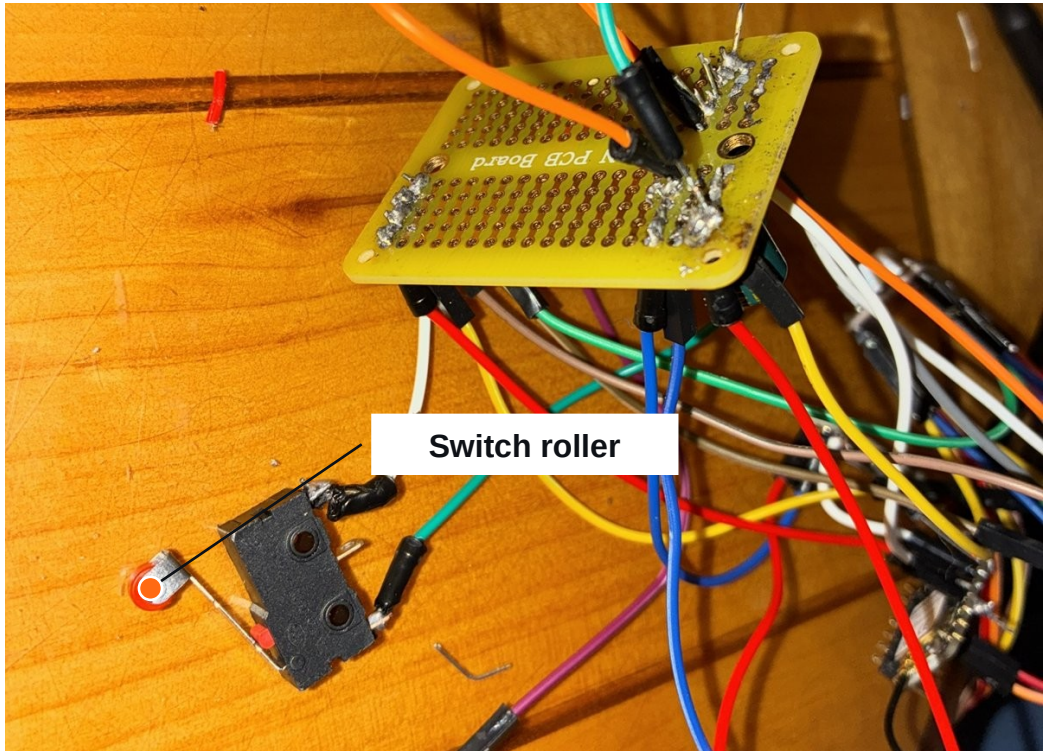
FUNDAMENTALS

$$r(n) = r_0 + n \cdot d \cdot p; \quad T = \tau / r(n); \quad \ell = 2\pi r$$

20 ft of 0.5 mm line on the Ø15 core: 5 layers, r = 7.5 to 10 mm, T = 23.8 to 17.9 N, line per rev 47 to 63 mm (+33 %).

CALCULATED

Bead must never cross the pulley; rev C missed by 5 mm.



PLANNED — not yet run

Trip test: a 5 mm bead on real line against the printed rev D arm.

1	WANTED	Bead on the line trips a home switch.
2	REALIZED	Mid-span, the bead rides over the pulley.
3	DECIDED	A drop arm under the pulley.
4	RESULT	**Rev C missed by 5 mm; rev D overlaps 3.75 mm.**
5	BUT	Never trip-tested on real line yet.

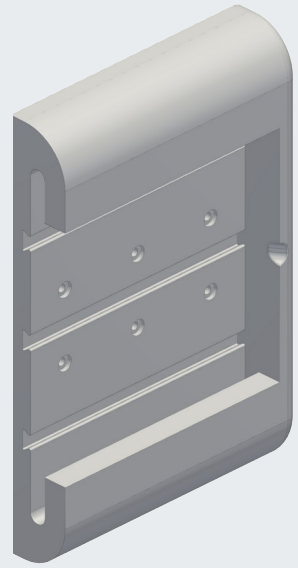
FUNDAMENTALS

$$Y\ \text{overlap} = (r_{bead} + r_{roller}) - Y_{roller}$$

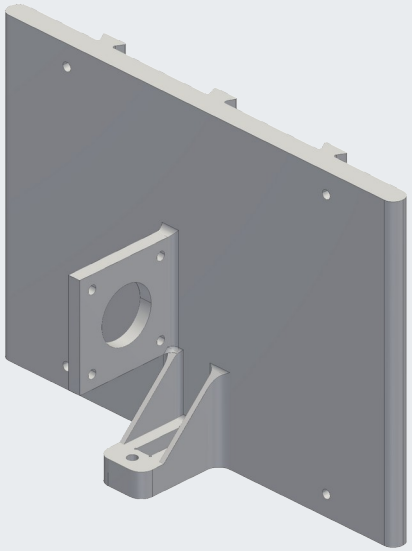
(2.5 + 2.25) − 1.0 = 3.75 mm, need ≥ 2.0 mm. X error ≤ 1.0 mm across 18 to 22 mm pulleys, 0 after the ±2 mm slot.

PLANNED

A removable station moves the anchor from a screw to a slide.



Fixed wall frame



Slide-in housing

PLANNED — result after the Oct 22 test

SolidWorks linear static study (stress, safety factor, deflection) and a physical station pull test.

1	WANTED	Swap a station without re-calibrating the room.
2	REALIZED	Anchor error maps 1:1; studs see pull-out.
3	DECIDED	Fixed frame + slide-in; SF 3 at 34 N, 1.5 at 68 N.
4	RESULT	**SolidWorks static study pending.**
5	BUT	Statics can't see creep; pull test too.

FUNDAMENTALS

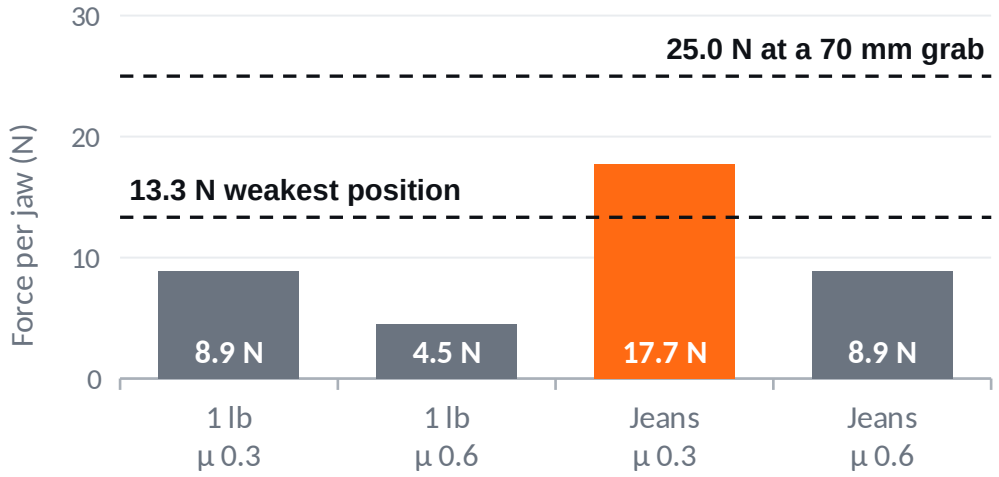
$$\sigma = M\,c\,/\,I \text{ (rail root)};\quad \delta = F\,L^3\,/\,(3\,E\,I) \text{ (cable exit)}$$

Pass: SF ≥ 3 at the 34 N slip load ($\sigma \le 8.33$ MPa on 25 MPa derated PETG) and SF ≥ 1.5 at 68 N dynamic; ≤ 0.5 mm at the cable exit.

CALCULATED
SIMULATED

A clamp stalls the servo every grip, so I sized for stall.

Bars: needed per jaw · Dashed lines: jaw force available



1 **WANTED** Five soft tendon fingers.

2 **REALIZED** A clamp is simpler but stalls the servo.

3 **DECIDED** Module 1.5, 2:1 ratio, my own 36T gear.

4 **RESULT** Gears 157 g to 96 g; 257 checks pass.

5 **BUT** **Jeans at μ 0.3 need 17.7 N; jaws give 13.3–25 N.**



My v6 compound, 3:1

gear set 157 g (CAD)



Rebuilt, 2:1

gear set 96 g (CAD)

FUNDAMENTALS

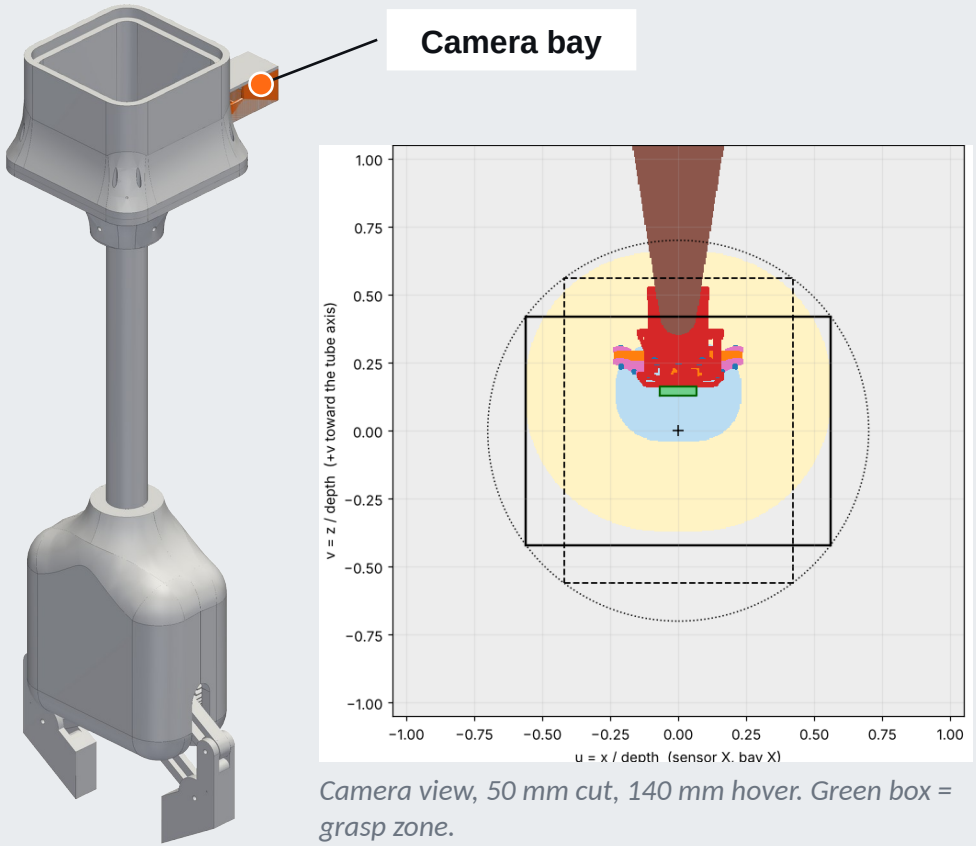
$$\sigma = F_t / (b \cdot m \cdot Y); \quad N \geq m(g + a) / (2\mu)$$

Module 1: SF 0.35–0.41 at stall; module 1.5, 16 mm face: SF 2.0–2.5. Jeans: 0.9 × 11.81 / (2 × 0.3) = 17.7 N per jaw.

CALCULATED

SIMULATED

The camera can't see the grasp at contact, and that's fine.



Camera view, 50 mm cut, 140 mm hover. Green box = grasp zone.

1	WANTED	Long aluminium arm; camera sees the jaws.
2	REALIZED	At contact, a plate hides the grasp spot.
3	DECIDED	Cut 50 mm of tube, not 100 mm.
4	RESULT	**Visible from 140 mm hover (was 172 mm); saves 33 g.**
5	BUT	Last ~14 cm is blind; add a hover correction.

FUNDAMENTALS

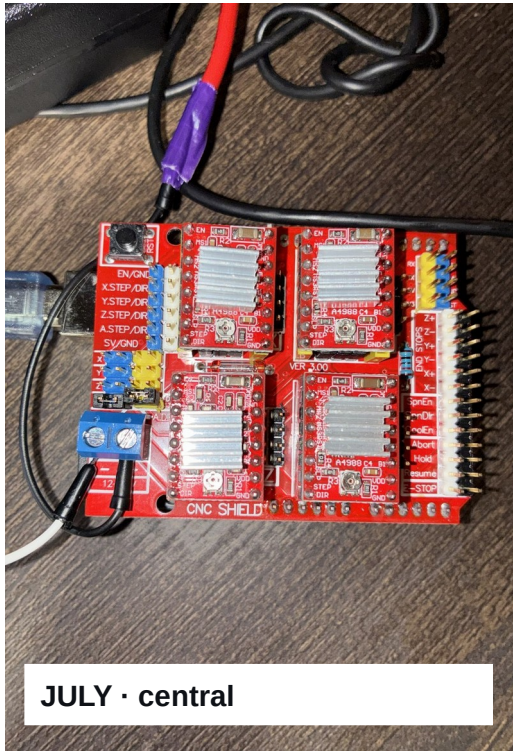
$$D^* = (k\,(H_0 - c) - 93) \,/\, (1 - k), \quad k = 1 - 59.5/97$$

$k = 0.387$, $H_0 = 512$ mm: $c = 0$ gives $D^* = 171$ mm, $c = 50$ mm gives 140 mm; each mm cut lowers D^* by 0.63 mm.

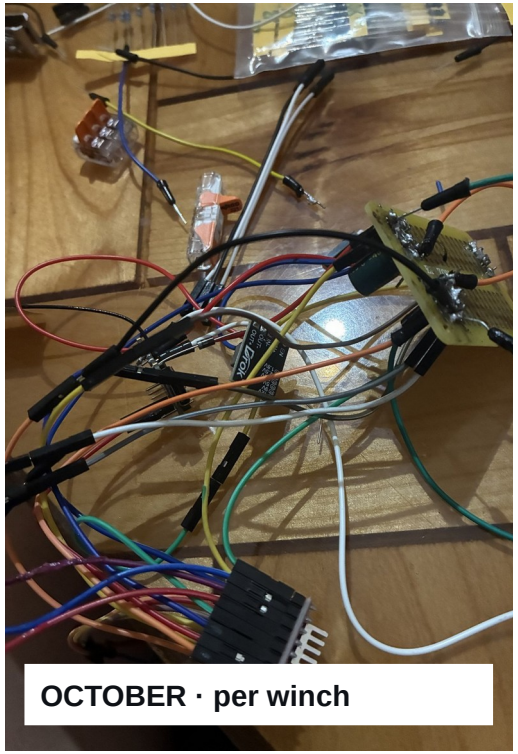
CALCULATED

SIMULATED

One controller per winch fixed the wiring and priced sync error.



JULY · central



OCTOBER · per winch

PLANNED — not yet run

Four stations, 1 lb, 300 mm at 50 mm/s: pass if each cable is within 2 mm.

1	WANTED	July: one controller; synchronised motion came free.
2	REALIZED	32 ft harnesses; skew becomes force on four cables.
3	DECIDED	**Host plans; stations run timed segments, own clocks.**
4	RESULT	Control logic simulated only.
5	BUT	Not yet run on four stations.

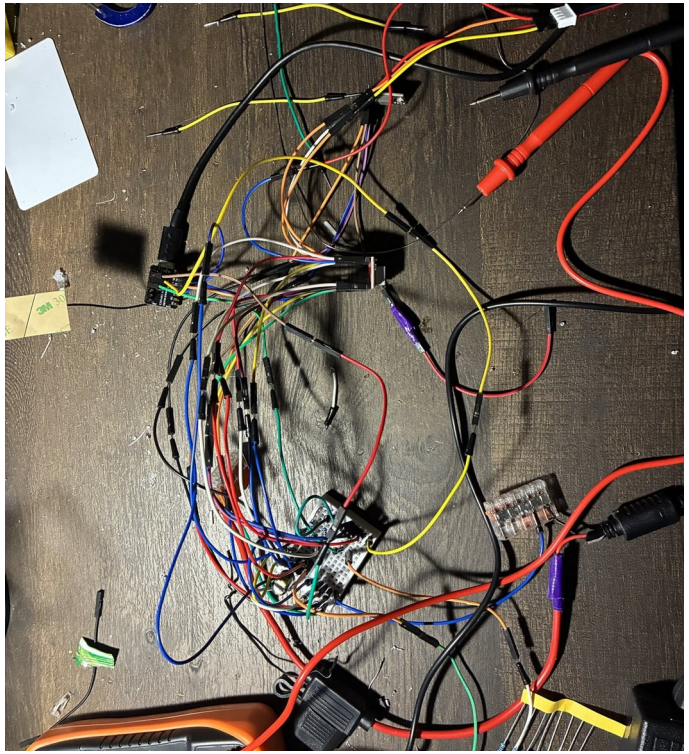
FUNDAMENTALS

$$\Delta\ell = v\cdot\Delta t;\quad F = k\cdot\Delta\ell;\quad V = I\cdot\rho\cdot 2L / A$$

50 mm/s × 50 ms = 2.5 mm; 3 N/mm × 2.5 mm = 7.5 N. 16 AWG, 32 ft, 2 A: 0.51 V drop (18 AWG: 0.82 V).

MEASURED

A motor that wouldn't move taught me to measure before coding.



3.3 V
expected on the logic rail

1.14 V
read on the meter

1	WANTED	First motion on the new station board.
2	REALIZED	**Pulses sent, shaft still; logic rail read 1.14 V.**
3	DECIDED	Solder 3.3 V; pull resistors; JP1; measure VDD.
4	RESULT	Rev D board, net-checked; motion after repair.
5	BUT	As-built board still needs its meter checklist.

PLANNED — not yet run

Meter checklist on the as-built board: VDD, SLEEP, ENABLE and the buck output.

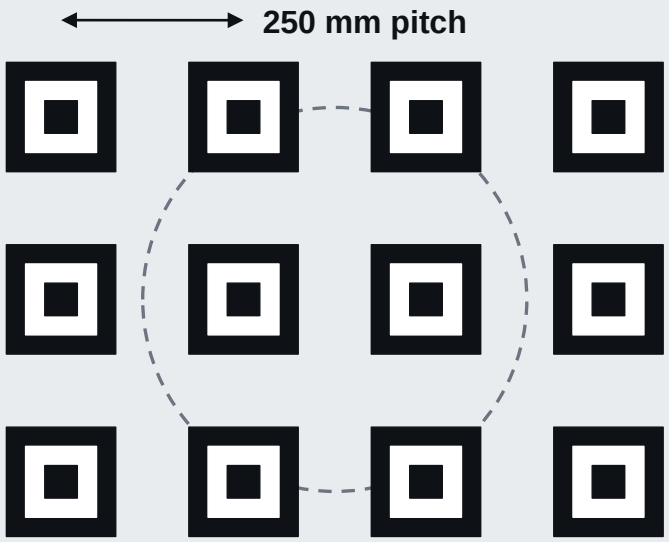
FUNDAMENTALS

$$I = V_{ref} / (8 \cdot R_s) = 0.85\ V / 0.8\ \Omega = 1.06\ A$$

Meter first: logic rail 1.14 V read against 3.3 V expected. The same habit on Oct 21 found the Vref pot topping out at 0.85 V.

CALCULATED
SIMULATED

I stopped buying precision and moved the safety into margins.



6.6 mm → 0.7 mm

lever-arm error per frame at 0.51 m: one tag vs the four-tag board

1	WANTED	Absolute cable lengths from vision alone.
2	REALIZED	One tag errs 6.6 mm; σ > 12 mm overtravel.
3	DECIDED	**Margins carry safety; probe slack; drop the IMU.**
4	RESULT	Simulated: 6/6 startups; 3 of 6 reach full trust.
5	BUT	Claw tilts 13–21° near corners; all simulated.

FUNDAMENTALS

$$\delta x = \sigma_\theta \cdot L; \quad \sigma_N = \sigma / \sqrt{N}$$

One tag: 12.9 mrad × 0.51 m = 6.6 mm. Four-tag board: 1.3 mrad × 0.51 m = 0.68 mm. 91–97 % of slack readings resolved (sim).

Measured, calculated, simulated, planned: the test that decides the rest.

MEASURED	CALCULATED	SIMULATED	PLANNED, with pass limits
• Vref cap 0.75–0.85 V • One motor on all four driver channels • 18 V on the meter at the supply • Buck output 5.00 V • Fin fits the motor • Logic rail 1.14 V (the fault)	• Cable statics, 32 to 4.6 N • Gear stress (Lewis) • Fin efficiency, 74 % to 37 % • Wire drop, 0.51 V • Grip force per jaw	• Control system and sync • Startup calibration • Camera view vs tube length • Jaw force, four-bar model	• Hold test: motor case below 60 °C • Station pull test, empty and full spool • SolidWorks station study: SF 3 at 34 N, 1.5 at 68 N, ≤ 0.5 mm • 1 lb, four stations, 300 mm at 50 mm/s: ≤ 2 mm per cable

FIRST TEST

Luggage-scale pull at 1.06 A, empty and full spool. Under 25 N at a full spool means a 3:1 reduction.

WHAT I DELETED, WHAT I LEARNED

My best decisions were deletions.

DELETED

- The IMU
- The yaw guard
- Five fingers, now two jaws
- 50 mm of tube
- The housing block, now an open frame (−600 g)
- The 5 lb reserve: a 1 lb gate first

THREE LESSONS

01 **Measure before you trust a margin**

The 0.85 V Vref cap and the 1.14 V rail

02 **Print the coupon before the part**

The spool bore: Ø5.2 became Ø5.45

03 **Stall is the design case**

Gear stall and motor holding heat, not the average

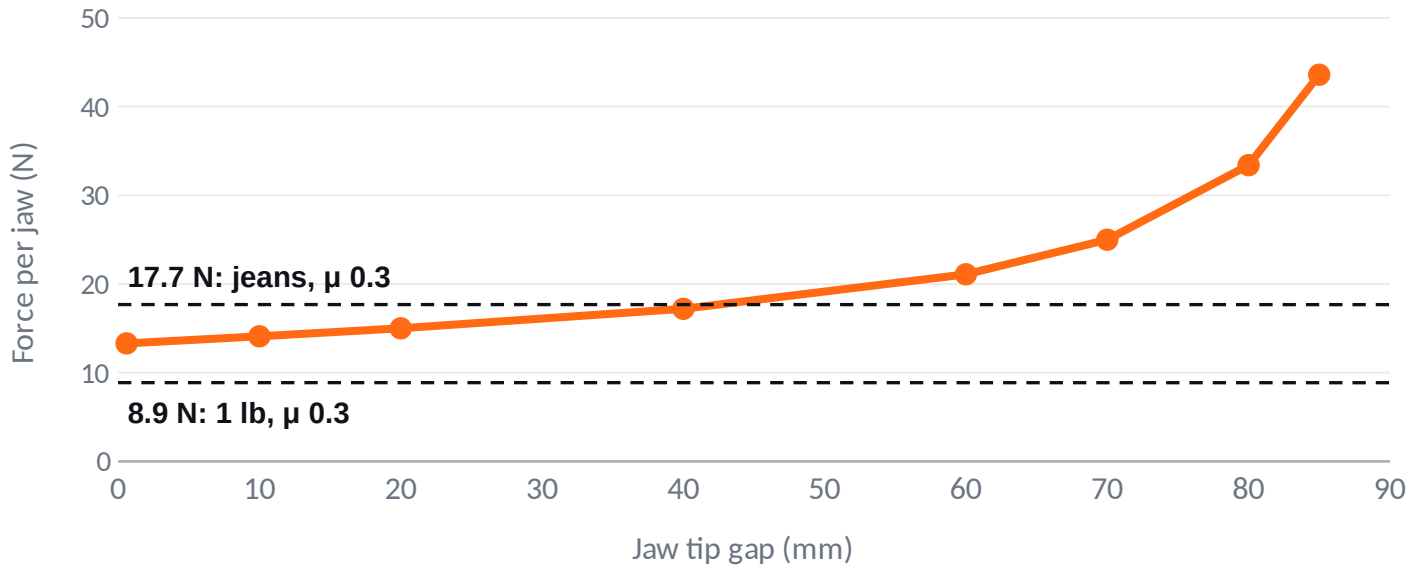
With half the time again: buy pulleys and couplers sooner, and build one station to test before four.

jasper_buntinx@tamu.edu

SIMULATED

CALCULATED

Jaw force is lowest closed and over 3× higher near full open.



76.5°
jaw stroke from a 153° servo (2:1)

91.6 mm
open gap, closed gap 0.000 mm

0.83 N·m
jaw torque at 4.8 V stall

Near full open the four-bar nears toggle: the effective lever arm shrinks (59 mm at a 10 mm gap, 33 mm at 70 mm) and force rises. No toggle inside the stroke.

FUNDAMENTALS

$$F_{jaw} = \tau_{jaw} / r_{eff};\quad \tau_{jaw} = N \cdot \tau_{in} \cdot \eta^2 / 2$$

$\tau_{jaw} = 2 \times 0.922\ \text{N·m} \times 0.95^2 / 2 = 0.83\ \text{N·m}$. $r_{eff} = 59, 48, 33$ mm at 10, 40, 70 mm gap, so F = 14.1, 17.2, 25.0 N.

SIMULATED

The host plans; each station executes, and stop never disables.

HOST

Planner

Plans the path and sends timed segments

timed segments

STATION 1

Own clock

feed-hold · lease

STATION 2

Own clock

feed-hold · lease

STATION 3

Own clock

feed-hold · lease

STATION 4

Own clock

feed-hold · lease

≈ 18 ms

stop spread across stations (simulated)

10 / 10

lossy motions completed at 2–5 % packet loss (sim)

Feed-hold in path time

All four stations ramp to zero along the same geometric path.

Lease

No valid host message in time: the station holds and tells its peers.

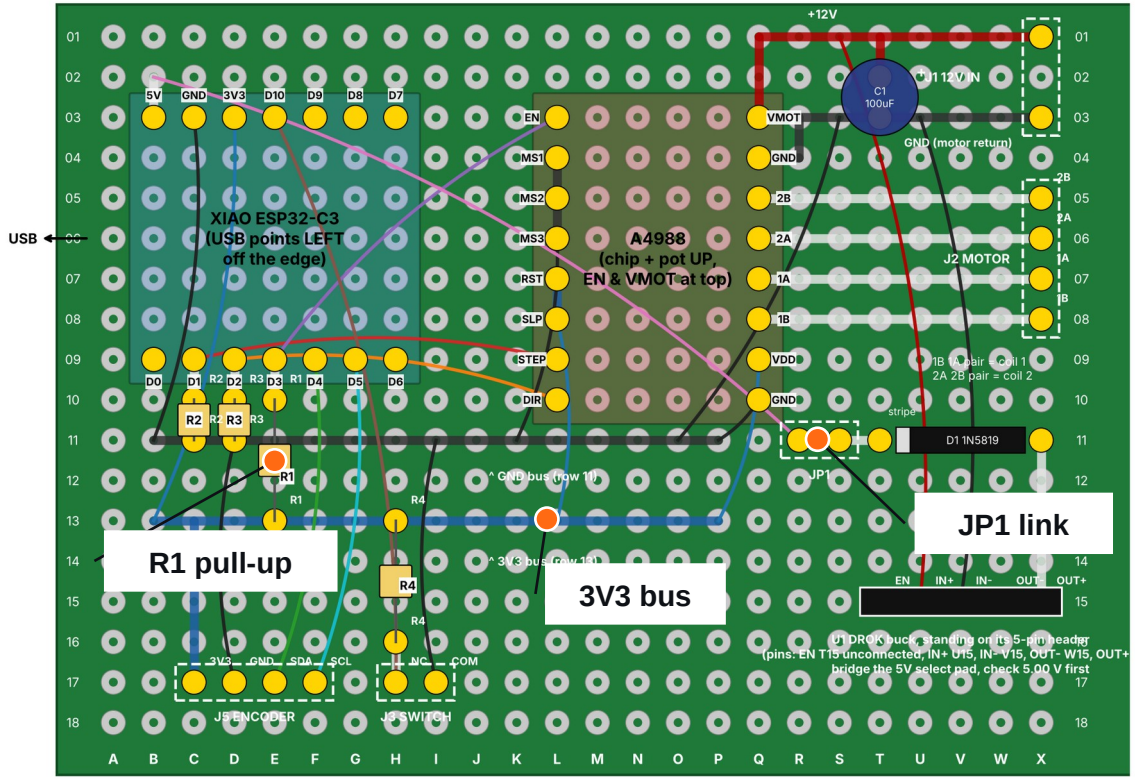
Stop never disables

Faults and pause hold torque. Only a confirmed disarm releases the load.

CALCULATED

PLANNED

Rev D: soldered 3.3 V, driver boots disabled, no USB-buck fight.



COMPONENT SIDE VIEW. Thick grey/red/blue = bare wire or solder run on the bottom. Thin coloured = insulated wire on the bottom (crossings OK). Yellow = soldered hole.

Match module pin LABELS to the holes before soldering headers: XIAO 5V at B03, D0 at B09; A4988 EN at L03, VMOT at Q03, DIR at L10.

WHAT REV D CHANGED

3.3 V soldered, not jumpered.

10 k pull resistors: the driver boots disabled.

JP1 and diode D1: USB and the buck never fight.

Net-checked: every hole has one net, every net is connected.

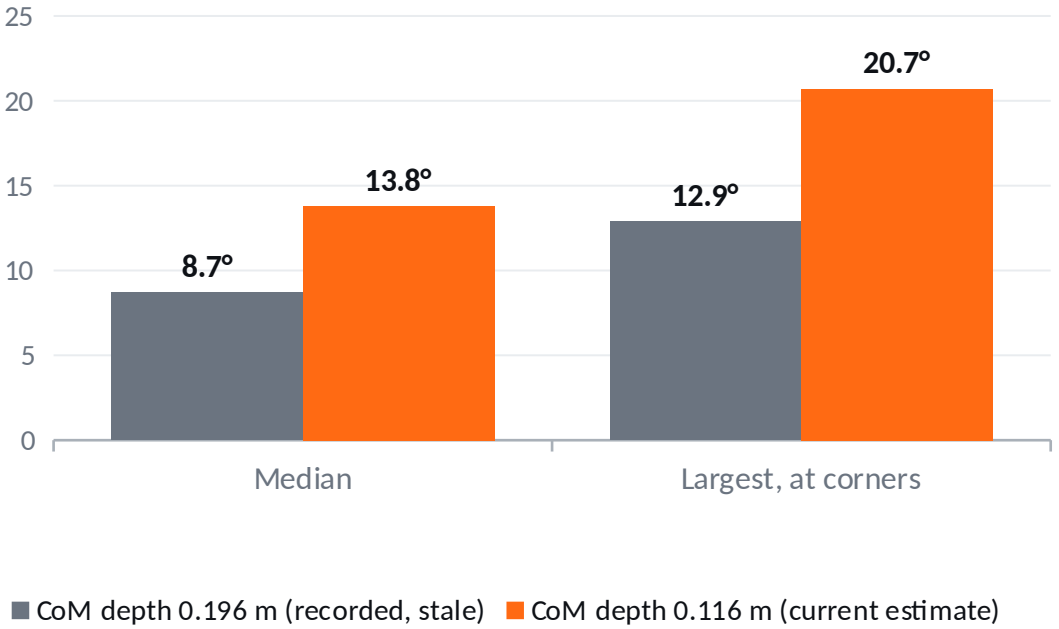
PLANNED — result after the Oct 22 test

Meter checklist on the as-built board: VDD, SLEEP, ENABLE, buck output.

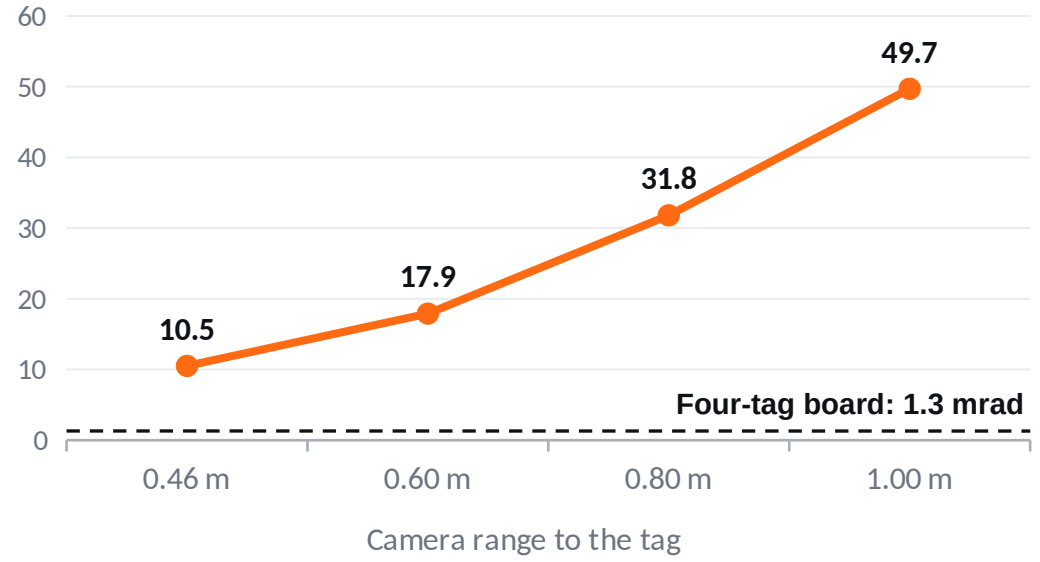
CALCULATED
SIMULATED

A floor board cuts tilt noise 10×; hanging tilt still reaches 13–21°.

HANGING TILT AT GRID POINTS (°)



TILT NOISE, ONE TAG, VS RANGE (mrad)



FUNDAMENTALS

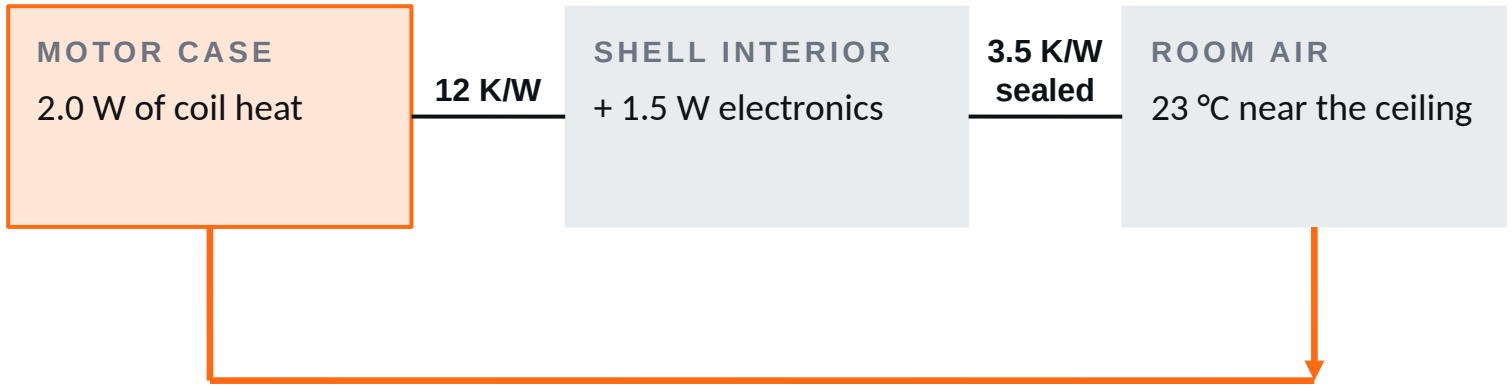
$$\theta \approx M_v / (W \cdot L); \quad \delta x = \sigma_\theta \cdot L$$

Corner tilt grows 1.61× when L falls from 0.196 to 0.116 m (1/L law: 1.70×); 83 % then 92 % of grid points exceed 5°. Board: 0.68 mm vs 6.6 mm for one tag.

CALCULATED

PLANNED

The fin is a parallel heat path; PETG alone can't carry it.



Aluminium fin: 22.1 K/W as built, 10.9 K/W as drawn

Sealed shell, 1.06 A: 64 °C with no fin, 46 °C with the as-built fin, 40 °C with the drawn 1/8 in fin.

PETG VS ALUMINIUM

	Aluminium	PETG
k, W/m·K	160	0.2
Flange path, K/W	0.011	14.0
Heat spreads, mm	74	2.6

PLANNED — result after the Oct 22 test

Fin FEA after the SolidWorks study; the hold test replaces the lumped model.

FUNDAMENTALS

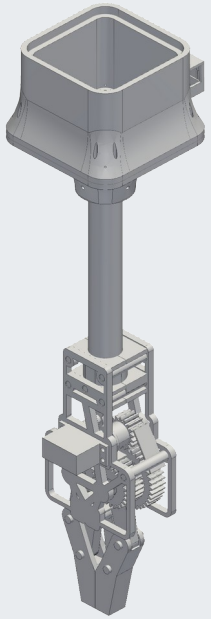
$$P_m = (T_m - T_{room}) / R_{fin} + (T_m - T_{shell}) / R_{ma}$$

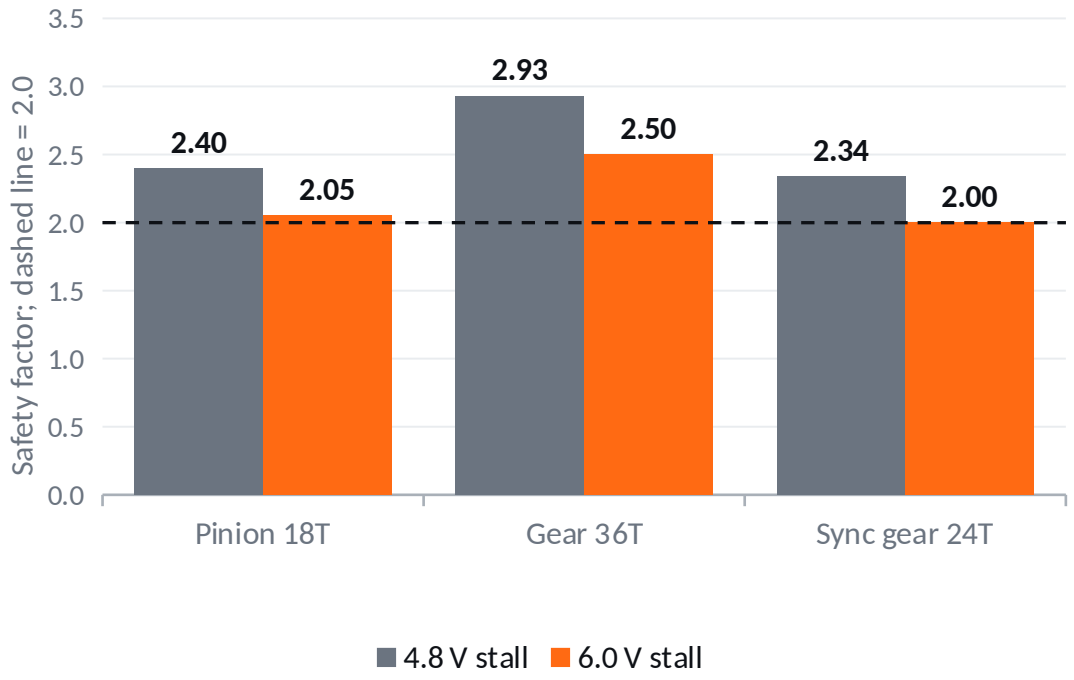
Aluminium 160 vs PETG 0.2 W/m·K gives $\sqrt{(160 / 0.2)} = 28\times$ at equal geometry; the flange path alone is 1260× worse. Sealed-shell time constant 35 min.

CALCULATED

SIMULATED

Open-frame v7 gearbox holds SF of about 2 or better at stall.





VERIFIED (REV 2)

Frame 143.5 g (limit 150 g).

Printed total 332 g; head 481 g.

72 checks pass; no interference over the sweep.

Open: servo ear pitch 47.0 vs 48.3 mm.

FUNDAMENTALS

$$SF = \sigma_{allow} / \sigma; \quad \sigma = F_t / (b \cdot m \cdot Y)$$

σ_{allow} = 22.5 MPa. At 6 V stall F_t = 79.9 N: pinion 11.0 MPa, SF 2.05. Sync gear, one jaw takes all: F_t = 113.9 N, SF 1.997.

Safety: one hard stop, studs only, nobody underneath.



One hard stop

The switched power strip is the only stop. Faults and pause hold torque.



Studs only

Stations anchor into wall studs; the line pulls them off the wall, so screws see pull-out.



Nobody under it

Nobody stands under the robot while it moves.



Balance charging

Charge the 2S LiPo with a matched balance charger, pack off the robot.

No safety-rated function is claimed (ISO 13849 and IEC 61508 are not claimed).